



Use HAPI for Easy Data Access

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Data Servers	Access Method
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CDAWeb	HAPI
	Custom

SSCWeb	HAPI
	Custom

INTER-MAGNET	HAPI
	Custom

VirES / SWARM	HAPI
	Custom

HAPI offers a single way to obtain data from many repositories

HAPI standardizes key data access pain points:

1. Same way to request the data.
2. Same arrangement (data structures) of returned data.

Python code to get data is short

```
# example showing how to get OMNIWeb data
from hapticlient import hapi

server      = 'https://cdaweb.gsfc.nasa.gov/hapi'
dataset     = 'OMNI2_H0_MRG1HR'
start       = '2021-10-25T00:00:00Z'
stop        = '2021-12-01T00:00:00Z'
# parameters is a comma-separated list
parameters  = 'DST1800,Proton_QI1800'

# Configuration options for the hapi function.
opts = {'logging': True, 'usecache': True, 'cachedir': './hapticache' }

# Get parameter data
data, meta = hapi(server, dataset, parameters, start, stop, **opts)
```

Python code to get different data is nearly the same

```
# example showing how to get OMNIWeb data
from hapiclient import hapi

server      = 'https://cdaweb.gsfc.nasa.gov/hapi'
dataset     = 'OMNI2_H0_MRG1HR'
start       = '2021-10-25T00:00:00Z'
stop        = '2021-12-01T00:00:00Z'
# parameters is a comma-separated list
parameters  = 'DST1800,Proton_QI1800'

# Configuration options for the hapi function.
opts = {'logging': True, 'usecache': True, 'cachedir': './hapicache' }

# Get parameter data
data, meta = hapi(server, dataset, parameters, start, stop, **opts)
```

Python code to get different data is nearly the same

```
# example showing how to get OMNIWeb data
from hapiclient import hapi

server      = 'https://cdaweb.gsfc.nasa.gov/hapi'
dataset     = 'RBSP-A_DENSITY_EMFISIS-L4'
start       = '2021-10-25T00:00:00Z'
stop        = '2021-12-01T00:00:00Z'
# parameters is a comma-separated list
parameters  = 'density'

# Configuration options for the hapi function.
opts = {'logging': True, 'usecache': True, 'cachedir': './hapicache' }

# Get parameter data
data, meta = hapi(server, dataset, parameters, start, stop, **opts)
```

Many Datasets Now Available!

HAPI Server	# Datasets
WDC	5,309
CDAWeb	3,578
SuperMAG	3,270
Intermagnet	3,073
CSA	1,989
SWHV	1,446
AMDA	1,067
KNMI	1,003
CCMC	319
SSCWeb	313
VirES for SWARM	174
Helioviewer	79
UCLA	44
LISIRD	29
NOAA / NCEI	25
Das2	19
Madrigal	15
Total	21,752

Try out HAPI using the HAPI Data Explorer

`http://hapi-server.org/servers`

Steps to take:

For “HAPI servers” : select CDAWeb

For “Datasets”: select RBSP-A_DENSITY_EMFISIS-L4

For “Parameters”: select “electron density”

For the time range: the defaults are ok

For output: select Data (can also select Image, or Script)

For format: select CSV

Data request for EMFISIS



https://cdaweb.gsfc.nasa.gov/hapi/data?id=RBSP-A_DENSITY_EMFISIS-L4¶meters=density&time.min=2012-09-15T00:00:03Z&time.max=2012-09-17T00:00:03.000Z



Visit the tables!

- More about HAPI data that is available.
- HAPI tutorials in HelioCloud
- Help with specific questions!